# Digital Content Verification System

## Fake News Detection using Deep Learning (LSTM)

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# Introduction

- **Context:** The digital information age has exponentially increased the velocity and volume of news dissemination.
- **Threat:** Proliferation of *misinformation* (Fake News) undermines social stability and computational trust.
- **Challenge:** Manual fact-checking is unscalable. Misinformation utilizes sophisticated linguistic markers (e.g., targeted sensationalism) that evade basic filters.
- **Objective:** Develop a robust Deep Learning framework for real-time veracity classification via sequential text analysis.

# Problem Statement

**The Research Gap**

Standard statistical models often rely on word frequencies (bag-of-words), failing to capture the **sequential context** and **narrative flow** essential to identifying deceptive content.

**System Requirements:**

1. Model long-term sequential dependencies in text data.
2. Maximize *Precision* to mitigate the False Positive rate (i.e., incorrectly flagging legitimate journalism).
3. Ensure production-ready deployment via a user-friendly interface.

# Dataset & Dimensionality Reduction

- **Source:** WELFake Dataset ($n = 72,134$ articles).
- **Target:** Binary classification ($0 =$ REAL, $1 =$ FAKE).

**Preprocessing Pipeline:**

- Lowercasing & Regex noise reduction.
- NLTK stop-word filtration.
- Sequence Padding (300 tokens max).

**Data Splitting (80/10/10):**

- Train: $57,707$ samples.
- Val: $7,213$ samples.
- Test: $7,214$ samples.

# Methodology: Classical Baseline

**Logistic Regression + TF-IDF**

- **Feature Extraction:** TF-IDF (Term Frequency-Inverse Document Frequency).
- **Constraint:** Bag-of-words approach; fundamentally ignores local syntax and word order.
- **Dimensionality:** Vectorized to the top $10,000$ features.

| Metric | Result |
|---|---|
| Accuracy | 95.77% |
| Recall | 96.08% |

Table: Baseline performance results on test set.

# Methodology: Proposed LSTM Model

**Long Short-Term Memory (LSTM) Architecture**

- **Embedding Layer:** Maps 15,000 unique tokens into a 128-dimensional continuous vector space.
- **Recurrent Layer:** 64 LSTM hidden units utilizing internal gates (Forget, Input, Output) to retain critical context.
- **Regularization:** Dropout and recurrent dropout set at $p = 0.2$.

$$\text{Output} = \sigma(Wx + b) = \frac{1}{1 + e^{-(Wx+b)}} \tag{0.1}$$

**Training Protocol:** Adam optimizer, Binary Cross-Entropy loss, trained over 3 epochs with a batch size of 128.

# Aggregate Results on Test Set

| Model | Acc | Prec | Rec | F1 |
|---|---|---|---|---|
| Logistic Regression | 95.77% | 95.63% | **96.08%** | 95.86% |
| **LSTM Network** | 94.01% | **97.12%** | 90.93% | 93.92% |

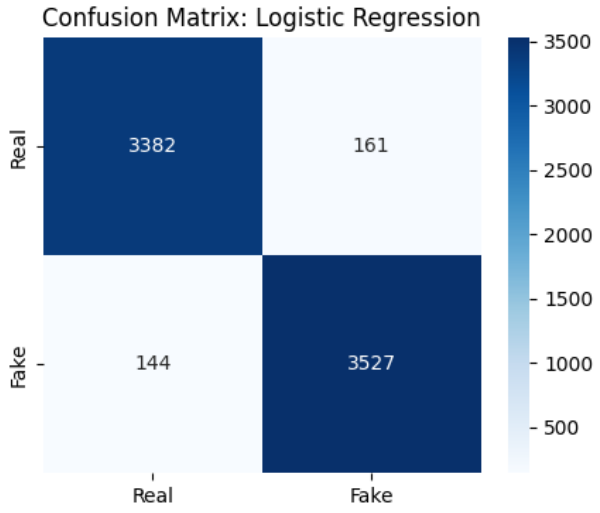
Table: Performance on unseen test set ($n = 7,214$).

**Detailed LSTM Classification Report:**

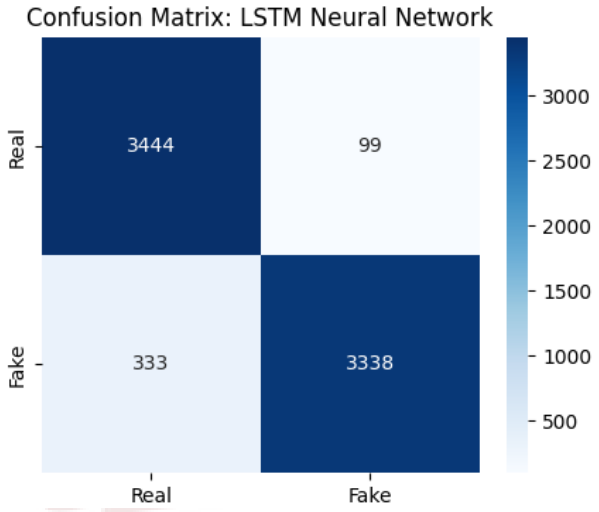
- **Class 1 (Fake News):** Precision of **97%**.
- **Class 0 (Real News):** Recall of **97%**.

# Model Justification: The Precision Priority

## Logistic Regression




## LSTM Network




### Why deploy the LSTM?

While the baseline has a marginally higher accuracy, the **LSTM minimizes False Positives (higher precision)**. Ethically, in content verification, a model must be highly conservative before classifying legitimate journalism as fabricated. Furthermore, the LSTM captures the sequential narrative buildup of misinformation.

# Deployment & Live Inference

- **Frontend UI:** Interactive Gradio Dashboard.
- **Input Methods:**
  - **Direct Input:** Manual article pasting.
  - **URL Scraping:** Live extraction via BeautifulSoup4.
  - **File Upload:** Bulk processing of .txt documents.
- **Cloud Infrastructure:** Hosted on Hugging Face Spaces with automated GitHub Actions (CI/CD) integration.

**Live Test Validation:** Successfully flagged a fabricated NASA article with an output confidence probability of **99.29%**.

# Conclusion and Future Work

**Summary:**

- Developed a robust, sequentially-aware LSTM framework.
- Achieved a highly reliable **97.12% precision rate**, bridging the gap between Deep Learning research and production.

**Future Directions:**

1. **Attention Mechanisms:** Transitioning to Transformer architectures (BERT/RoBERTa) for bidirectional context.
2. **Multi-lingual:** Expanding the tokenizer to accommodate French and Wolof news corpora.
3. **Multi-modal Integration:** Incorporating CNNs for image forensics and deepfake verification.

# Key References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